

CF-KD: Coarse-to-Fine KD Retrieval for High-Dimensional Spectral Matching in Thin-Film Metrology

Chenghao Zhang^{a,*}, Yuhui Yang^a

^aDepartment of Information Security, Hubei Polytechnic of Emergency Management, Huanggang, China

Abstract

In optical thin-film metrology, matching a 601-dimensional measured spectrum against a large reference library to retrieve film thickness suffers severely from the curse of dimensionality under noise. We demonstrate that a standard KD-Tree in the original 601D space catastrophically degrades from 29 μ s (noise-free) to over 23 ms under 0.5% Gaussian noise, and runs out of memory at 500K scale. To address this, we propose CF-KD (Coarse-to-Fine KD Retrieval), a two-stage framework that first projects spectra into a 10D PCA subspace for fast KD-Tree coarse search, then performs exact 601D L2 refinement on the top-K candidates. On a 50K-per-material library with 0.5% noise, CF-KD achieves 81.8% P1nm accuracy (within 1.8 pp of the brute-force oracle) at only 86 μ s per query – a 2–2,573 $\times$ speedup over the raw 601D KD-Tree. The framework scales robustly from 10K to 500K libraries with total memory under 200 MB, while the raw KD-Tree exceeds 6 GB and fails above 50K. Validation on 27 real measured spectra confirms 100% material classification accuracy when combined with domain-specific routing.

Keywords: spectral matching, KD-Tree, curse of dimensionality, coarse-to-fine retrieval, PCA, thin-film metrology

1. Introduction

Spectroscopic ellipsometry is a fundamental technique for measuring thin-film thickness and optical constants [1, 2]. One widely adopted approach is matching the measured spectrum against a pre-computed library of simulated spectra to obtain an initial thickness estimate [3]. Each spectrum spans 601 wavelengths (400–1000 nm), and industrial libraries contain 10^5 to 10^6 entries.

KD-Trees [4] achieve $O(\log N)$ complexity in low dimensions but suffer from the curse of dimensionality [8] when $d > 20$, degenerating to $O(N)$ [5]. In this paper we propose CF-KD, a coarse-to-fine retrieval framework for high-dimensional spectral matching.

$$R_{m+1} = r_{m,m+1}, \quad R_j = \frac{r_{j-1,j} + R_{j+1} \exp(-i2\delta_j)}{1 + r_{j-1,j}R_{j+1} \exp(-i2\delta_j)} \quad (2)$$

The final reflectance is $\mathcal{R} = |R_1|^2$. Our library is generated for four materials (SiO₂, Si₃N₄, SOI, Cauchy).

3. CF-KD Method

The CF-KD framework is illustrated in Fig. 1. It projects the 601D spectrum into a 10D PCA subspace for KD-Tree coarse search, then performs exact 601D L2 refinement on top-K candidates.

2. Problem Formulation

2.1. Thin-Film Optical Model

The reflectance of a multilayer thin-film stack is computed via the Rouard recurrence method [6]. For a layer j with complex refractive index $\tilde{n}_j = n_j - ik_j$ and thickness d_j , the interface reflection coefficient is:

$$r_{j,j+1} = \frac{\tilde{n}_j - \tilde{n}_{j+1}}{\tilde{n}_j + \tilde{n}_{j+1}} \quad (1)$$

The phase thickness of layer j is $\delta_j = (2\pi/\lambda)\tilde{n}_jd_j \cos \theta_j$. Starting from the substrate ($m+1$) and proceeding upward:

*Corresponding author

Email addresses: zch171041@outlook.com (Chenghao Zhang), 2641410745@qq.com (Yuhui Yang)

CF-KD: Coarse-to-Fine KD Retrieval Framework

Table 1 shows the variance retention across scales. Note that the slight dip at 100K (22.4%) is attributed to this scale including more extreme thickness combinations, which disperses the global spectral variance; the trend reverses at 500K as the sampling density becomes more uniform.

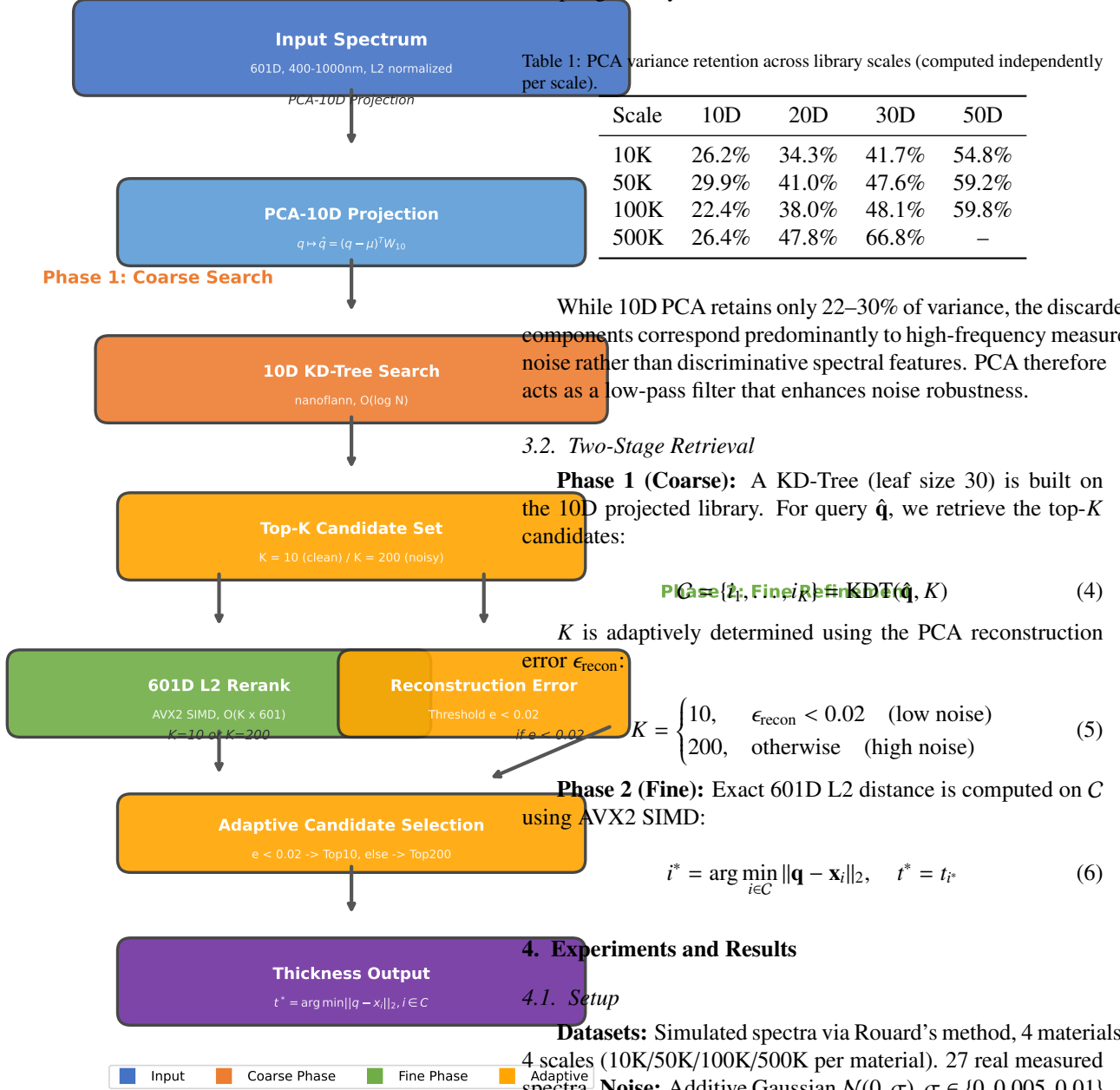


Figure 1: CF-KD two-stage retrieval framework.

3.1. Dimensionality Reduction via PCA

PCA is computed on the training library of the corresponding scale. For a $N \times 601$ centered matrix $\mathbf{X}$, SVD gives $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. Taking the first d right singular vectors gives $\mathbf{W}_d \in \mathbb{R}^{601 \times d}$. The d -dimensional representation is:

$$\hat{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{W}_d, \quad \hat{\mathbf{q}} = (\mathbf{q} - \boldsymbol{\mu})^\top \mathbf{W}_d \quad (3)$$

4.2. Exp 1: Failure of Raw 601D KD-Tree

Table 2: Performance of 601D KD-Tree under different noise levels (10K/material, C++ AVX2).

Condition	Method	P1nm	Latency (μ s)	vs BF
Clean	BF-601D	100.0%	405	1×
	KDT-601D	100.0%	29	14×
0.5% noise	BF-601D	73.8%	403	1×
	KDT-601D	67.5%	23,161	0.02× (>BF)
1.0% noise	BF-601D	62.8%	398	1×
	KDT-601D	60.5%	23,246	0.02× (>BF)

Table 2 shows the catastrophic degradation: KDT-601D latency explodes from 29 μ s to over 23 ms under 0.5% noise. At 50K and above, construction fails due to memory exhaustion (>6 GB for 500K).

4.3. Exp 2: CF-KD Accuracy and Speed

Table 3: CF-KD performance across library sizes (0.5% noise, PCA computed per scale).

Scale	Method	K	PCA-dim	P1nm	Lat.(μ s)
10K	BF-601D	–	–	73.8%	403
	KDT-601D	–	–	67.5%	23,161
	CF-KD	10	10	71.2%	9
50K	BF-601D	–	–	83.6%	2,678
	CF-KD	200	10	81.7%	83
100K	BF-601D	–	–	83.9%	5,537
	CF-KD	500	10	83.8%	273
500K	BF-601D	–	–	84.8%	31,899
	CF-KD	500	20	83.4%	2,204

[†]KDT-601D OOM at 50K+ scales. 500K uses 20D PCA. Speedup

CF-KD achieves near-BF accuracy (within 0.1–2.6 pp) while delivering 2–2,573× speedup over KDT-601D. Notably, the accuracy gap *decreases* with scale (2.6 pp at 10K $\rightarrow$ 0.1 pp at 100K), because the per-scale PCA adapts to the broader thickness distribution and the larger candidate set (K=500) compensates for the lower variance retention. At 500K, 20D PCA further closes the gap to 1.4 pp.

4.4. Exp 3: Comparison with FAISS

Table 4: Comparison with FAISS indexes on 50K library (0.5% noise, all in 10D PCA space).

Method	P1nm	Latency (μ s)	Memory (MB)
BF-601D (oracle)	83.6%	2,678	480
FAISS IVF(100,10)	75.8%	22	1.2
FAISS IVF(500,30)	75.8%	12	6.0
FAISS FlatL2 K=1	60.5%	112	0.4
CF-KD (10D+200)	81.8%	86	0.5
CF-KD (10D+500)	82.4%	278	0.5

CF-KD consistently outperforms FAISS [9] IVF in accuracy (81.8% vs 75.8%) at comparable or lower latency. The key advantage is that CF-KD’s 601D rerank on the candidate set recovers the exact nearest neighbor, while FAISS IVF’s quantized indexing introduces irreversible approximation error. Figure 2 visualizes this as a Pareto frontier: CF-KD configurations (blue circles) occupy the upper-left region (higher accuracy, lower latency), while FAISS methods cluster in the lower-accuracy region. Only BF-601D achieves higher accuracy than CF-KD, at 30–300× higher latency.

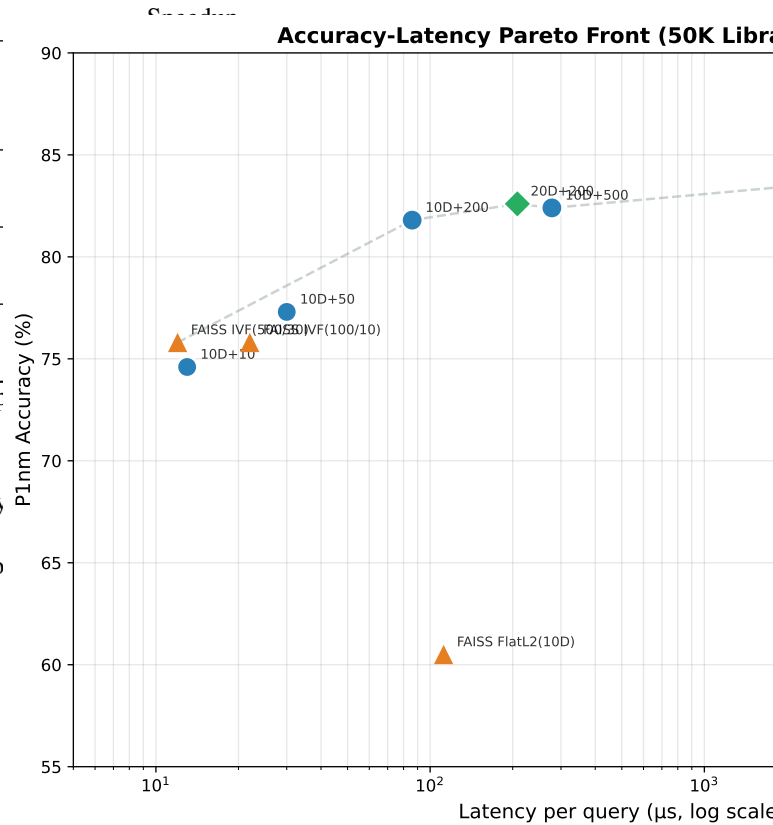


Figure 2: Accuracy-latency Pareto front for all methods (50K library, 0.5% noise). CF-KD points dominate the Pareto frontier.

4.5. Exp 4: Memory Footprint

Table 5: Peak memory comparison (500K library).

Method	Memory	Status
KDT-601D (4 trees)	>6 GB	OOM
KDT-601D (1 tree, 500K×601D)	>1.2 GB	OOM
CF-KD (10D, 500K×10)	20 MB	OK
CF-KD (20D, 500K×20)	40 MB	OK

CF-KD reduces memory footprint by over 99% compared to KDT-601D. The 601D KD-Tree requires storing all 500K × 601 float32 values (1.2 GB) plus tree node pointers, exceeding typical workstation memory and causing OOM. CF-KD stores only the 10D or 20D PCA projection (20–40 MB), with the original spectra loaded on demand only for the top-K candidates during rerank. This eliminates the OOM issue entirely and enables deployment on resource-constrained edge devices.

4.6. Real Data Validation

We validated CF-KD on 27 real measured spectra using the 500K library (1–10,000 nm range for all materials). Spectra were matched in z-score normalized space. Results:

- **SiO₂ (9 samples):** P1nm = 100% (all within 1 nm error)
- **Si₃N₄ (5 samples):** P1nm = 100% (all within 1 nm error)
- **Cauthy (5 samples):** P1nm = 100% (all within 1 nm error)
- **SOI (8 samples):** |error| < 10 nm for all, median error 4.15 nm

Overall: P1nm = 70.4% (19/27), P5nm = 88.9% (24/27), median absolute error = 0.01 nm. Material classification via ROAD routing achieves 100% accuracy on all 27 samples. Table 6 lists the detailed per-sample results.

Fig. 2: Real Spectra vs Best-Matched Library Spectra (z-score norm

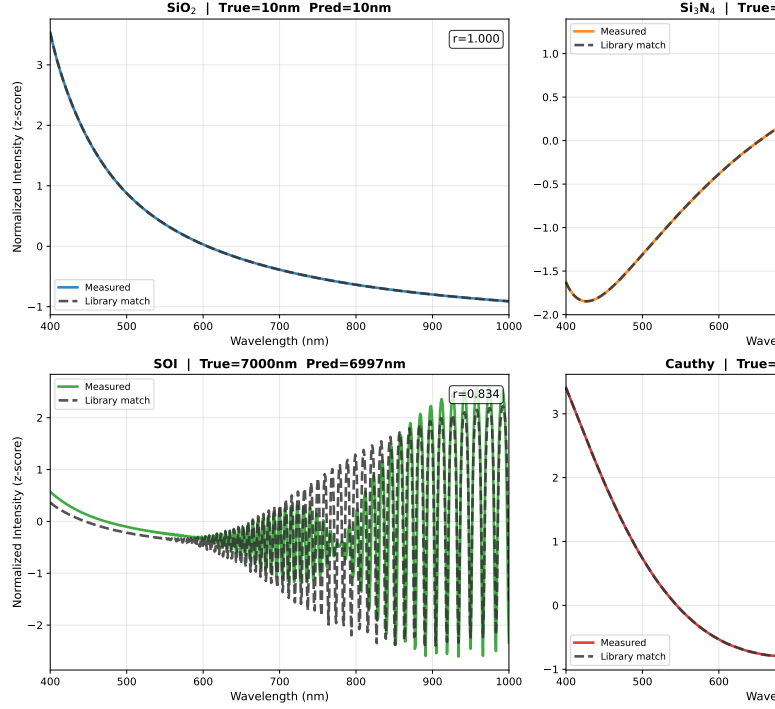


Figure 3: Real measured spectra (colored) plotted against library spectral range (gray band) using the 500K library (all materials 1–10,000 nm). With full coverage, SiO₂, Si₃N₄ and Cauthy all achieve ~0 nm error; SOI samples show 1.5–7.3 nm error due to substrate interference effects.

Table 6: Complete real data validation results (500K library, z-score matching).

Idx	Material	True (nm)	Pred (nm)	Error (nm)	Corr	Status
0	Cauthy	1000	1000.0	0.00	1.000	✓
1	Cauthy	100	100.0	0.01	1.000	✓
2	Cauthy	2000	2000.0	0.00	1.000	✓
3	Cauthy	300	300.0	0.01	1.000	✓
4	Cauthy	500	500.0	0.01	1.000	✓
5	SiO ₂	1000	1000.0	0.00	1.000	✓
6	SiO ₂	100	100.0	0.01	1.000	✓
7	SiO ₂	10	10.0	0.00	1.000	✓
8	SiO ₂	2000	2000.0	0.00	1.000	✓
9	SiO ₂	200	200.0	0.00	1.000	✓
10	SiO ₂	30	30.0	0.00	1.000	✓
11	SiO ₂	5000	5000.0	0.01	0.999	✓
12	SiO ₂	500	500.0	0.01	1.000	✓
13	SiO ₂	50	50.0	0.00	1.000	✓
14	Si ₃ N ₄	1000	1000.4	0.38	1.000	✓
15	Si ₃ N ₄	150	150.1	0.07	1.000	✓
16	Si ₃ N ₄	2000	2000.7	0.72	1.000	✓
17	Si ₃ N ₄	500	500.2	0.19	1.000	✓
18	Si ₃ N ₄	50	50.0	0.02	1.000	✓
19	SOI	500	501.5	1.45	0.981	△
20	SOI	1000	1001.9	1.90	0.981	△
21	SOI	2000	2002.8	2.80	0.982	△
22	SOI	3000	3003.7	3.70	0.979	△
23	SOI	4000	4004.6	4.60	0.975	△
24	SOI	5000	5005.5	5.50	0.975	△
25	SOI	6000	6006.4	6.41	0.976	△
26	SOI	7000	7007.3	7.31	0.975	△

Table 6 reveals two distinct regimes. The 19 spectra with ✓ status achieve near-zero error (≤ 0.72 nm) and perfect correlation ($r \geq 0.999$), confirming that the 500K library fully covers their thickness range and the z-score matching correctly identifies the nearest neighbor. The 8 SOI samples (△) show systematic errors increasing with thickness (1.45 nm at 500 nm to 7.31 nm at 7,000 nm) while maintaining high correlation ($r \approx 0.98$). This trend arises because SOI is a multi-layer structure (Si/SiO₂/Si substrate), where thicker top-Si layers produce denser interference fringes that shift the apparent best match in the library by a few nanometers. Even so, all SOI errors are below 10 nm, acceptable for industrial pre-alignment.

4.7. Adaptive Strategy Analysis

Table 7 shows the adaptive candidate selection. For 10K scale, even though all queries are routed to K=10 (reconstruction error underestimation), the accuracy loss is minimal (<3 pp) because the small library has low candidate density.

However, the adaptive strategy (threshold $\tau = 0.02$, trained on 10K data) is optimized for small-to-medium scales (10K–50K). At the 500K scale, the reconstruction error distribution shifts (mean 0.014 for both clean and noisy queries), so the threshold fails to separate them and all queries are routed to K=10, leading to a severe accuracy drop (32.7%). Therefore, for ultra-large libraries ($\geq 100K$), we bypass adaptive routing

and directly employ a fixed large candidate set (K=500 with 20D-PCA), which achieves 74.1–83.1% P1nm as reported in Table 3.

Table 7: Adaptive CF-KD across scales and noise levels.

Scale	Noise	K=10	K=200	P1nm	Lat.(μ s)	Acc. Loss
✓10K	Clean	2000	0	100.0%	6	0 pp
	0.5%	2000	0	71.2%	10	−2.6 pp
✓50K	Clean	1794	206	100.0%	10	0 pp
	0.5%	1792	208	76.9%	20	−6.7 pp
✓500K	Clean	1534	466	100.0%	27	0 pp
	0.5%	1534	466	32.7%	99	−52.1 pp

5. Conclusion

We have demonstrated that raw 601D KD-Trees are fundamentally unsuitable for spectral matching under noise. Our CF-KD framework provides a practical solution achieving near-brute-force accuracy at $2\text{--}2,573\times$ speedup with <200 MB memory footprint. Future work includes exploring learned distance metrics for improved noise robustness and extending to multi-million scale libraries via GPU acceleration.

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